Darjeeling City Transport Plan

City: Darjeeling

AI-Generated Sustainable Transport Plan

01 July 2026

Executive Summary

The city of Darjeeling faces significant transport challenges, including high private vehicle usage and inadequate public transport. The recommended plan involves implementing a Bus Rapid Transit (BRT) system on the Siliguri-Darjeeling corridor and a cycling network on the Darjeeling-Ghum corridor. However, the plan has several weaknesses, including a lack of data on key metrics and uncertainty in financial projections. The overall verdict is that the plan needs revision, with further studies and re-evaluation of the recommended modes required. The revised plan aims to provide a more efficient, sustainable, and integrated transport system for the city.

Approach

The plan was developed using a multi-agent method, involving data analysis, design, and review. The approach considered ridership and cost models to recommend the most suitable transport modes for each corridor. However, the lack of data on key metrics and uncertainty in financial projections have impacted the plan's robustness. Further studies and revisions are needed to strengthen the plan.

1. Data Foundation (Agent 1)

The plan is grounded in 505 researched facts drawn from 34 sources (Wikipedia, government mobility plans, traffic studies).

Key sources

- Road traffic statistics
- [PDF] Market Assessment for Intercity Electric Buses in India
- [PDF] SECTORAL INSIGHTS: TRANSPORT - NITI Aayog
- Pedaling On: Celebrating One Year of ITDP's Cycling Cities Campaign - Institute for Transportation and Development Policy
- [PDF] Equitable Public Bus Network Optimization for Social Good
- Darjeeling's new Himalayan cycle trail | India holidays | The Guardian
- Darjeeling Local Sightseeing Car Fare (2026) | Taxi Price Tips
- How To Travel Around Sikkim in Shared Vehicles? - Krishnandu Sarkar

2. Demand Analysis (Agent 2)

Key metrics

- Population Current: 2,070,000 (2021)
- Population Growth Rate: not specified
- Estimated Daily Trips: not specified
- Current Mode Share: private vehicles: 53% (2025), public transport: 26.34% (Scenario a), walking: 31%, cycling: 6%
- Existing Pt Capacity: not specified
- Pt Demand Gap: not specified

Mobility patterns

mode share: private vehicles: 53% (2025), public transport: 26.34% (Scenario a), walking: 31%, cycling: 6%; peak patterns: 11% to 17% of daily passengers travel during the peak hour (7:00-8:00 a.m.); key od pairs: Siliguri-Darjeeling, Darjeeling-Ghum; commute times: not specified

Current demand

total daily trips: not specified; current pt capacity vs demand: not specified; underserved: not specified

Future demand

growth rate: 3.7% CAGR for India's overall passenger transport demand up to 2030; projected population trips trip length: population: 2,160,000 (2026), trips: not specified, trip length: 6.26 km (Scenario a), 4.45 km (Scenario b); motorization trend: increasing, with private vehicle ownership reaching 197 vehicles per 1,000 persons in 2023

Capacity gaps

trips that cannot be served: not specified; areas with no pt: not specified; worst times: peak hour (7:00-8:00 a.m.)

Public-transport deficiencies

coverage: not specified; frequency: not specified; reliability: not specified; comfort gaps: not specified

Bottlenecks

- location: Hill Cart Road; reason: congestion due to high traffic volume; severity: not specified

3. Proposed Solutions (Agent 3)

Siliguri-Darjeeling

Siliguri -> Darjeeling | length 51.2 km, 51 stops (length: Nominatim geocode + haversine x1.3 detour)

Mode	Daily Rid.	Peak/hr	Capacity	Capex (cr)	Break-even	Verdict
CYCLING	26,250	3,150	5,000	80.64	Never (subsidy)	VIABLE
BRT	42,000	5,040	12,000	1919.1	136.7 yr	NOT VIABLE
METRO	105,000	12,600	45,000	13740.0	Never (subsidy)	NOT VIABLE

Recommended: BRT

Why: BRT is recommended as its peak capacity of 12000 pphpd comfortably covers the peak-hour demand of 5040, and it is the cheapest mode that meets this demand. Metro is oversized with a peak capacity of 45000 pphpd, exceeding the demand, and has a higher capital cost. Cycling has a lower peak capacity of 5000 pphpd, which is insufficient for the demand.

Alternatives rejected: Metro is rejected due to its high capital cost and oversized capacity. Cycling is rejected due to its insufficient peak capacity.

Darjeeling-Ghum

Darjeeling -> Ghum | length 4.4 km, 4 stops (length: Nominatim geocode + haversine x1.3 detour)

Mode	Daily Rid.	Peak/hr	Capacity	Capex (cr)	Break-even	Verdict
CYCLING	4,550	546	5,000	6.93	Never (subsidy)	VIABLE
BRT	7,280	874	12,000	167.7	52.1 yr	NOT VIABLE
METRO	18,200	2,184	45,000	1205.0	Never (subsidy)	NOT VIABLE

Recommended: CYCLING

Why: Cycling is recommended as the primary mode for this short corridor, given its low peak-hour demand and short route length. BRT and metro are oversized and have higher capital costs, making them less suitable for this corridor.

Alternatives rejected: BRT and metro are rejected due to their high capital costs and oversized capacities.

Citywide recommendations

Implement a citywide cycling network as a feeder system to complement the BRT and metro lines. Consider phasing the implementation of BRT to manage costs and assess demand more accurately. Conduct further studies on population growth, daily trips, and existing transport capacity to inform the plan.

Assumptions & caveats

- Population growth rate and estimated daily trips may affect demand projections.
- Existing public transport capacity may impact the viability of the recommended modes.
- Financial projections for BRT in the Siliguri-Darjeeling corridor are subject to uncertainty.
- Alternative scenarios for demand and financial projections should be considered to strengthen the plan's robustness.

4. Independent Review (Agent 4)

Verdict: NEEDS_REVISION

The plan has several weaknesses, including a lack of data on key metrics such as population growth rate, estimated daily trips, and existing public transport capacity. The recommended modes for the corridors may not be the most suitable, and the financial projections are subject to uncertainty. Further studies and revisions are needed to strengthen the plan.

Key weaknesses identified

- lack of data on key metrics
- uncertainty in financial projections
- oversized peak capacity of BRT mode
- high capital cost of BRT mode
- unrealistic break-even period of BRT mode

Improvement suggestions

- conduct further studies on population growth rate, estimated daily trips, and existing public transport capacity
- re-evaluate the recommended modes for the corridors based on the results of the further studies
- consider alternative modes, such as bus rapid transit with a lower peak capacity
- provide more information on how the recommended modes will be funded, maintained, and integrated with the existing transport network

Conclusion

In conclusion, the Darjeeling City Transport Plan requires revision to address its weaknesses and provide a more effective transport system for the city. The recommended next steps include conducting further studies on population growth rate, estimated daily trips, and existing public transport capacity. The plan should be re-evaluated based on the results of these studies, and alternative modes should be considered. By revising the plan, the city can develop a more efficient, sustainable, and integrated transport system that meets the needs of its citizens.